
Cross-Language Probe Invariance Depends on Readout Choice

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Cross-language probe transfer is often interpreted as evidence that a code model rep-
2 represents program semantics independently of programming language. We test this
3 interpretation for variable roles in Qwen2.5-Coder-1.5B and StarCoder2-7B using
4 parallel Python, JavaScript, and PHP solutions. A masked character n -gram classi-
5 fier loses 0.126 macro-F1 on transfers involving Python relative to JavaScript $\leftrightarrow$ PHP.
6 A residual-stream probe differs by only -0.012 (95% problem-clustered interval
7 $[-0.034, 0.020]$), which appears to support language-invariant role encoding. This
8 probe mean-pools the occurrence’s own tokens, whereas the surface classifier masks
9 the identifier. Excluding the occurrence from the readout reduces mean transfer
10 from 0.844 to 0.589 and changes the boundary effect to -0.060 $[-0.088, -0.030]$.
11 The shift is -0.048 $[-0.088, -0.016]$. Context-matched untrained networks show
12 boundaries of -0.064 and -0.035 across two weight initializations. Neither com-
13 parison supports a reliably flatter trained boundary. StarCoder2-7B reproduces
14 the reversal on the same items with a larger separation, moving from -0.014
15 $[-0.036, 0.018]$ span-pooled to -0.147 $[-0.186, -0.092]$ occurrence-excluded.
16 Its trained-minus-untrained boundary difference is -0.068 $[-0.114, -0.009]$, so
17 its trained boundary is reliably steeper. Thus the original invariance is not identified
18 independently of the readout, while the restricted readout still contains training-
19 dependent role information.

20 1 Introduction

21 Linear probes measure whether a label is decodable from a representation, not whether the model
22 uses it, and their performance can reflect probe capacity or information already available in the
23 probe input [Hewitt and Liang, 2019, Voita and Titov, 2020, Ravichander et al., 2021, Elazar et al.,
24 2021, Belinkov, 2022]. Cross-language transfer adds a further interpretation. If a probe fitted on one
25 programming language transfers to another, the decoded feature may appear independent of surface
26 form.

27 Existing work has recovered code syntax across languages with probe baselines [Hernández López
28 et al., 2026], separated language-specific from language-agnostic embedding components [Utpala
29 et al., 2024], and measured uneven alignment among programming languages on parallel pro-
30 grams [Kargaran et al., 2025]. Within-language suites map what code models encode across many
31 probe tasks [Karmakar and Robbes, 2021, Troshin and Chirkova, 2022, Karmakar and Robbes, 2024].
32 Our contribution is a comparator rather than a new task. These studies establish that code represen-
33 tations contain both shared and language-specific structure, but not a model-free, feature-restricted
34 comparator on the same cross-language prediction task.

35 We add that comparator for variable roles such as accumulator and index key [Sajaniemi, 2002].
36 Variable roles describe what a variable does, which makes them a plausible target for language-

independent representation. The initial probe result supports that interpretation. It changes when the readout no longer pools the occurrence tokens that carry the identifier, which parallel implementations often preserve [Kargaran et al., 2025]. This ablation localizes the instability to the readout support. It does not, by itself, identify lexical identity as the cause because pooling support and token position change together.

2 Setup

Corpus. XLCOST [Zhu et al., 2022] provides parallel solutions keyed by problem identifiers that are stable across languages. Python, JavaScript, and PHP are the only three languages with at least 500 shared problems for every pair. For example, C# and Java share only 175 problems (Appendix B). We first restrict each language to the 869 problems present in all three. Role extraction then determines which of those problems contain a usable occurrence for each role, so role-specific coverage still differs across cells. Data splits use a fixed hash of the problem identifier and never separate occurrences from the same problem.

The Python data retain formatting that the JavaScript and PHP data do not. Because this formatting difference coincides with the language contrast, we remove whitespace before constructing character features. Equalizing whitespace on both sides changes each of the three audited iterator transfers by less than 0.001.

Conditions. We evaluate four conditions on matched language pairs.

- **Surface n -gram.** We apply a character n -gram classifier to the context around an occurrence after replacing every instance of the variable’s name with a placeholder. The classifier is fitted on the source language and evaluated on the target. We report one fixed masked-window feature set rather than selecting the best feature set separately for each cell.
- **Probe, span-pooled.** Logistic regression on residual activations mean-pooled over the occurrence’s own tokens uses a layer chosen on source-language validation data. This readout includes the identifier tokens.
- **Probe, occurrence-excluded.** We use the same probe but pool the 16 tokens on either side of the occurrence while excluding its tokens. Model inputs and weights are unchanged. Only the readout is altered.
- **Untrained.** We evaluate the same architecture and tokenizer with randomly initialized weights under each pooling method. Randomly initialized networks can retain substantial lexical and structural information, so they provide a necessary floor [Zhang and Bowman, 2018, Hewitt and Liang, 2019]. Each probe is compared with the floor using the same readout.

Models are Qwen2.5-Coder-1.5B [Hui et al., 2024] and StarCoder2-7B [Lozhkov et al., 2024]. Their untrained controls use the corresponding architectures and tokenizers. Qwen2.5-Coder is continued-pretrained from Qwen2.5 [Qwen et al., 2024]. Reported uncertainty resamples complete problems rather than individual occurrences.

3 Surface transfer tracks the language contrast

The surface classifier reaches 0.929 between JavaScript and PHP without a model or identifier access, then falls to 0.804 on transfers involving Python. The table is an unweighted mean over cells. A separate pooled, problem-clustered estimate is -0.114 with a 95% interval of $[-0.147, -0.081]$. Effects vary across roles. Iterator contributes -0.249 , compared with -0.064 for accumulator and index_key together. Iterator labels have the weakest cross-language comparability because Python uses an AST-based extractor while JavaScript and PHP use regular expressions. Some cells contain only ten minority examples. We therefore treat the surface contrast as descriptive evidence for this extracted sample, not as a clean estimate of a language-family effect.

Table 1: Cross-language role transfer averaged over three roles. *Close* contains JavaScript $\leftrightarrow$ PHP. *Python* contains the four directions to or from Python. Probe floors use the corresponding readout.

	Qwen2.5-Coder-1.5B			StarCoder2-7B		
	close	Python	effect	close	Python	effect
Surface n -gram, name masked	0.929	0.804	-0.126	shared model-free baseline		
Probe, span-pooled (includes occurrence)	0.852	0.840	-0.012	0.877	0.863	-0.014
Probe, occurrence-excluded	0.630	0.569	-0.060	0.734	0.586	-0.147
untrained, occurrence-excluded	0.527	0.464	-0.064	0.552	0.473	-0.080

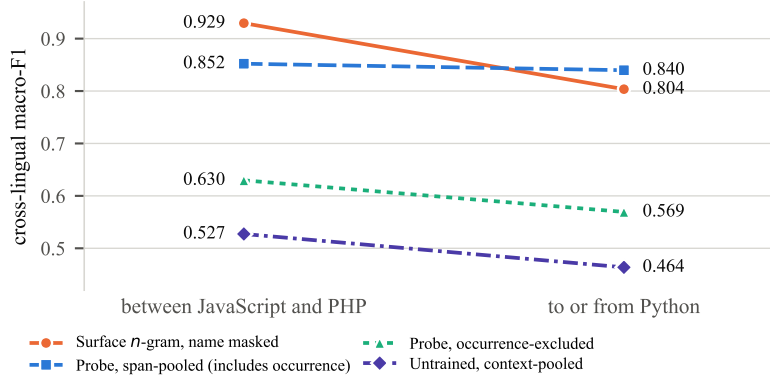


Figure 1: For Qwen2.5-Coder-1.5B, the surface baseline changes across the language contrast. The span-pooled probe appears flat but includes the occurrence tokens. Excluding the occurrence lowers transfer and yields a slope close to the matched untrained control.

83 4 The inferred boundary depends on readout support

84 The span-pooled probe reaches 0.844 overall. Its cross-language boundary is -0.012 with a problem-
85 clustered interval of $[-0.034, 0.020]$. This is the result that would normally be interpreted as
86 language-independent role decodability.

87 Excluding the occurrence from the readout lowers mean transfer from 0.844 to 0.589 (Figure 1). The
88 context-matched untrained model reaches 0.485. The boundary changes to -0.060 $[-0.088, -0.030]$.
89 The readout-induced shift is -0.048 $[-0.088, -0.016]$. The original flat boundary is therefore not
90 robust to where activations are read.

91 The occurrence-excluded untrained model has a -0.064 boundary $[-0.089, -0.039]$. Subtracting
92 this floor gives a trained-minus-untrained boundary of $+0.003$ $[-0.027, 0.036]$. A second weight
93 initialization yields a -0.035 $[-0.062, -0.008]$ floor boundary with an overall level of 0.497 against
94 the first initialization’s 0.485. The two floor levels are similar, while their boundary estimates
95 differ. The trained boundary (-0.060) lies within their range, at its steeper end. Against neither
96 initialization is it reliably flatter. Training still contributes. Trained-minus-untrained transfer is
97 $+0.102$ $[0.065, 0.132]$ close and $+0.106$ $[0.081, 0.127]$ for Python transfers against the first floor, and
98 $+0.109$ $[0.067, 0.145]$ and $+0.084$ $[0.058, 0.107]$ against the second.

99 **A second model.** StarCoder2-7B repeats the reversal on the same prediction items, more sharply.
100 Its span-pooled boundary is -0.014 $[-0.036, 0.018]$, as flat as Qwen’s. Excluding the occurrence
101 moves it to -0.147 $[-0.186, -0.092]$, a shift of -0.134 $[-0.168, -0.086]$ against Qwen’s -0.048 .
102 The models differ in how that boundary relates to its floor. StarCoder2’s untrained boundary is
103 -0.080 $[-0.110, -0.049]$. Its trained-minus-untrained difference is -0.068 $[-0.114, -0.009]$ and
104 excludes zero, whereas the same contrast was null in Qwen ($+0.003$ $[-0.027, 0.036]$). Readout
105 dependence appears in both tested models. Its relation to the untrained control differs between them.

106 Identifier exclusion also matters within languages. Mean in-language F1 falls from 0.898 to 0.771
107 against an untrained 0.623. Direct access to the occurrence accounts for a substantial part of both
108 in-language and cross-language decodability.

109 The remaining occurrence-excluded margin supports training-dependent role information in this
110 readout, but not a language-independent mechanism. In Qwen the training lift is similar on both sides
111 of the boundary. The surface baseline outperforms the context-pooled probe in 14 of 18 matched cells
112 with problem-clustered intervals below zero for the paired difference. This shows surface sufficiency
113 on the same prediction items. It is not a decomposition of the model signal, because the model still
114 processes unmasked source code.

115 5 Discussion

116 The central result concerns measurement. A probe that pools over an occurrence can use information
117 concentrated at those tokens, including lexical regularities preserved by parallel programs. A
118 randomly initialized model tests what training contributes under a fixed readout, not whether that
119 readout isolates the intended construct. Occurrence exclusion changes the pooled positions and their
120 information content together. Calling the difference “identifier leakage” would therefore overstate
121 what this experiment identifies.

122 The model-free condition exposes a different source of evidence. Its input withholds every target-
123 name occurrence, while the probe model sees the original program. The two predictors share labels
124 and test items but not information sets, so their comparison establishes surface sufficiency rather
125 than an isolated representational increment. Model-free baselines are established generally in code
126 probing [Troshin and Chirkova, 2022]. The cited cross-language studies do not fit one on the same
127 prediction items [Hernández López et al., 2026, Utpala et al., 2024, Kargaran et al., 2025]. The
128 present result extends the usual distinction between decodability and model use [Elazar et al., 2021,
129 Belinkov, 2022]. Decodability can also arise from a label correlate exposed directly by the readout.

130 **Reconciling with prior alignment results.** Kargaran et al. [2025] find Python the *least* aligned
131 of these three languages on parallel programs, whereas the span-pooled probe changes little across
132 transfers involving Python. The studies measure different quantities, so agreement is not required.
133 Even so, excluding the identifier moves the probe contrast toward their result, which is compatible
134 with a genuine Python boundary without establishing a common mechanism.

135 **Falsifying evidence.** The StarCoder2-7B experiment tests a prediction specified in advance. Its
136 close-pair margin over the matched control is +0.181, compared with Qwen’s +0.102, and its
137 boundary difference excludes zero. This limits the deflationary reading. In StarCoder2-7B, the
138 occurrence-excluded boundary is reliably steeper than its untrained counterpart. The measurement
139 claim remains unchanged because the boundary steepened rather than flattened. The interpretation
140 would weaken further if same-readout identifier replacement preserved the flat transfer curve, or if a
141 parallel corpus without shared naming conventions produced the original span-pooled invariance.

142 **Implication for linguistic medium studies.** When linguistic or symbolic form is manipulated,
143 the measurement instrument must not retain a feature the manipulation preserves. Parallel corpora
144 concentrate such features, including names, entities, and numbers. A feature-restricted model-free
145 comparator and a matched readout ablation are two inexpensive checks before invariance is attributed
146 to a representation.

147 **Limitations.** The study uses two models (1.5B and 7B), three languages, three roles, and one
148 context width. Python supplies the sole broad contrast. Role coverage varies after extraction, so
149 composition may affect group means. Iterator extractors differ across languages and carry most of
150 the surface effect. No multilingual hand-audit quantifies extractor error. The untrained controls use
151 two probe seeds rather than five and use two Qwen weight initializations but only one StarCoder2
152 initialization. The Qwen boundary varies across initializations (−0.064, −0.035), while its level does
153 not (0.485, 0.497). The cross-model comparison therefore rests on one StarCoder2 draw. Occurrence
154 exclusion may retain correlated nearby uses and changes pooling support rather than lexical identity
155 alone. These results concern decodability, not model use.

References

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a_00422.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359.
- José Antonio Hernández López, Martin Weyssow, Jesús Sánchez Cuadrado, and Houari Sahraoui. Syntactic multilingual probing of pre-trained language models of code. *Journal of Systems and Software*, 2026. doi: 10.1016/j.jss.2025.112604.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China, 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1275.
- Binyuan Hui, Jian Yang, Zeyu Cui, Jiayi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, Kai Dang, Yang Fan, Yichang Zhang, An Yang, Rui Men, Fei Huang, Bo Zheng, Yibo Miao, Shanghaoran Quan, Yunlong Feng, Xingzhang Ren, Xuancheng Ren, Jingren Zhou, and Junyang Lin. Qwen2.5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024. doi: 10.48550/arXiv.2409.12186.
- Amir Hossein Kargaran, Yihong Liu, François Yvon, and Hinrich Schuetze. How programming concepts and neurons are shared in code language models. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. doi: 10.18653/v1/2025.findings-acl.1379.
- Anjan Karmakar and Romain Robbes. What do pre-trained code models know about code? In *2021 36th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pages 1332–1336, Melbourne, Australia, 2021. IEEE. doi: 10.1109/ASE51524.2021.9678927.
- Anjan Karmakar and Romain Robbes. INSPECT: Intrinsic and systematic probing evaluation for code transformers. *IEEE Transactions on Software Engineering*, 50(2):220–238, 2024. doi: 10.1109/TSE.2023.3341624.
- Anton Lozhkov, Raymond Li, Loubna Ben Allal, Federico Cassano, Joel Lamy-Poirier, Nouamane Tazi, Ao Tang, Dmytro Pykhtar, Jiawei Liu, Yuxiang Wei, Tianyang Liu, Max Tian, Denis Kocetkov, Arthur Zucker, Younes Belkada, Zijian Wang, Qian Liu, Dmitry Abulkhanov, Indraneil Paul, Zhuang Li, Wen-Ding Li, Megan Risdal, Jia Li, Jian Zhu, Terry Yue Zhuo, Evgenii Zheltonozhskii, Nii Osa Osa Dade, Wenhao Yu, Lucas Krauss, Naman Jain, Yixuan Su, Xuanli He, Manan Dey, Edoardo Abati, Yekun Chai, Niklas Muennighoff, Xiangru Tang, Muhtasham Oblokulov, Christopher Akiki, Marc Marone, Chenghao Mou, Mayank Mishra, Alex Gu, Binyuan Hui, Tri Dao, Armel Zebaze, Olivier Dehaene, Nicolas Patry, Canwen Xu, Julian McAuley, Han Hu, Torsten Scholak, Sebastien Paquet, Jennifer Robinson, Carolyn Jane Anderson, Nicolas Chapados, Mostofa Patwary, Nima Tajbakhsh, Yacine Jernite, Carlos Munoz Ferrandis, Lingming Zhang, Sean Hughes, Thomas Wolf, Arjun Guha, Leandro von Werra, and Harm de Vries. StarCoder 2 and The Stack v2: The next generation. *arXiv preprint arXiv:2402.19173*, 2024. doi: 10.48550/arXiv.2402.19173.
- Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 3363–3377, Online, 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.eacl-main.295.

- 207 Jorma Sajaniemi. An empirical analysis of roles of variables in novice-level procedural programs. In
208 *Proceedings IEEE 2002 Symposia on Human Centric Computing Languages and Environments*,
209 pages 37–39, Arlington, VA, USA, 2002. IEEE Computer Society. doi: 10.1109/HCC.2002.
210 1046340.
- 211 Sergey Troshin and Nadezhda Chirkova. Probing pretrained models of source codes. In *Pro-*
212 *ceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks*
213 *for NLP*, pages 371–383, Abu Dhabi, United Arab Emirates (Hybrid), December 2022. As-
214 sociation for Computational Linguistics. doi: 10.18653/v1/2022.blackboxnlp-1.31. URL
215 <https://aclanthology.org/2022.blackboxnlp-1.31/>.
- 216 Saiteja Utpala, Alex Gu, and Pin-Yu Chen. Language agnostic code embeddings. In *Proceedings*
217 *of the 2024 Conference of the North American Chapter of the Association for Computational*
218 *Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024. doi: 10.18653/v1/
219 2024.naacl-long.38.
- 220 Elena Voita and Ivan Titov. Information-theoretic probing with minimum description length. In
221 *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*
222 *(EMNLP)*, 2020. doi: 10.18653/v1/2020.emnlp-main.14.
- 223 Kelly W. Zhang and Samuel R. Bowman. Language modeling teaches you more than translation
224 does: Lessons learned through auxiliary syntactic task analysis. In *Proceedings of the 2018*
225 *EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages
226 359–361, Brussels, Belgium, November 2018. Association for Computational Linguistics. doi:
227 10.18653/v1/W18-5448.
- 228 Ming Zhu, Aneesh Jain, Karthik Suresh, Roshan Ravindran, Sindhu Tipirneni, and Chandan K.
229 Reddy. XLCOST: A benchmark dataset for cross-lingual code intelligence. *arXiv preprint*
230 *arXiv:2206.08474*, 2022. doi: 10.48550/arXiv.2206.08474.

231 A Complete transfer results

Table 2: All eighteen transfer cells for Qwen2.5-Coder-1.5B. *Surface* is the strongest masked-context n -gram feature. *Name* uses the variable’s name alone. *Probe* is the residual-stream probe. Majority and shuffled-label controls sit near chance throughout. ρ is a descriptive one-instance score step, not a confidence interval or hypothesis test.

role	pair	n	surface	name	probe	major.	shuf.	ρ
accumulator	JavaScript→PHP	1,976	0.951	0.897	0.904	0.361	0.470	0.0005
accumulator	JavaScript→Python	3,341	0.813	0.873	0.875	0.415	0.473	0.0004
accumulator	PHP→JavaScript	1,903	0.954	0.897	0.918	0.372	0.490	0.0005
accumulator	PHP→Python	1,377	0.782	0.824	0.833	0.276	0.439	0.0008
accumulator	Python→JavaScript	3,294	0.785	0.874	0.918	0.370	0.504	0.0003
accumulator	Python→PHP	1,478	0.736	0.845	0.874	0.271	0.466	0.0007
index_key	JavaScript→PHP	1,976	0.940	0.834	0.897	0.384	0.478	0.0005
index_key	JavaScript→Python	3,341	0.859	0.787	0.883	0.330	0.497	0.0003
index_key	PHP→JavaScript	1,903	0.964	0.838	0.880	0.376	0.516	0.0005
index_key	PHP→Python	1,377	0.839	0.782	0.854	0.387	0.515	0.0008
index_key	Python→JavaScript	3,294	0.871	0.822	0.915	0.314	0.470	0.0003
index_key	Python→PHP	1,478	0.858	0.838	0.874	0.419	0.459	0.0008
iterator	JavaScript→PHP	1,976	0.937	0.862	0.906	0.448	0.488	0.0008
iterator	JavaScript→Python	3,341	0.774	0.733	0.939	0.444	0.472	0.0005
iterator	PHP→JavaScript	1,903	0.965	0.873	0.851	0.446	0.478	0.0008
iterator	PHP→Python	1,377	0.576	0.788	0.865	0.428	0.462	0.0010
iterator	Python→JavaScript	3,294	0.790	0.769	0.918	0.466	0.484	0.0007
iterator	Python→PHP	1,478	0.741	0.788	0.902	0.475	0.438	0.0020

Table 3: Problem-clustered intervals for every surface transfer cell. The final row is the Python-pair minus close-pair boundary effect.

role	pair	macro-F1	95% CI	problems
accumulator	JavaScript→PHP	0.925	[0.897, 0.946]	409
accumulator	JavaScript→Python	0.845	[0.812, 0.873]	389
accumulator	PHP→JavaScript	0.953	[0.929, 0.968]	409
accumulator	PHP→Python	0.844	[0.810, 0.873]	378
accumulator	Python→JavaScript	0.853	[0.815, 0.886]	389
accumulator	Python→PHP	0.861	[0.822, 0.891]	378
index_key	JavaScript→PHP	0.875	[0.823, 0.911]	409
index_key	JavaScript→Python	0.878	[0.836, 0.909]	389
index_key	PHP→JavaScript	0.931	[0.902, 0.954]	409
index_key	PHP→Python	0.846	[0.801, 0.882]	378
index_key	Python→JavaScript	0.887	[0.841, 0.919]	389
index_key	Python→PHP	0.881	[0.831, 0.915]	378
iterator	JavaScript→PHP	0.894	[0.808, 0.945]	409
iterator	JavaScript→Python	0.608	[0.554, 0.668]	389
iterator	PHP→JavaScript	0.972	[0.908, 0.994]	409
iterator	PHP→Python	0.564	[0.520, 0.621]	378
iterator	Python→JavaScript	0.848	[0.732, 0.916]	389
iterator	Python→PHP	0.813	[0.705, 0.891]	378
ALL	Python pairs – close pair	-0.114	[-0.147, -0.081]	432

Table 4: Probe transfer per model, averaged within each group. The untrained network shares Qwen2.5-1.5B’s architecture and tokenizer with randomly initialised weights. It is the floor a representational claim must clear, and it is itself flat.

model	close pair	Python pairs	all	vs. untrained
Qwen2.5-1.5B	0.899	0.888	0.892	+0.316
Qwen2.5-Coder-1.5B	0.893	0.887	0.889	+0.314
Qwen2.5-1.5B (untrained)	0.590	0.568	0.575	N/A

232 B Pairwise problem overlap in XLCoST

Table 5: Shared problem identifiers between every pair of XLCoST languages, train split. Matched cross-lingual transfer needs a pair to share enough problems to hold out a test fold. Only the three bold cells clear 500. No pair outside Python/JavaScript/PHP does, including same-family pairs.

	C++	C#	Java	JavaScript	PHP	Python
C++	N/A	115	122	75	17	74
C#	115	N/A	175	75	14	62
Java	122	175	N/A	85	11	66
JavaScript	75	75	85	N/A	1,529	2,953
PHP	17	14	11	1,529	N/A	1,145
Python	74	62	66	2,953	1,145	N/A

Table 6: Per-cell macro-F1 for the four intersection-sample conditions. *Lift* subtracts the context-matched randomly initialized result from the context-pooled trained result.

role	pair	surface	span	context	random	lift
accumulator	JavaScript→PHP	0.928	0.833	0.551	0.516	+0.035
accumulator	PHP→JavaScript	0.952	0.836	0.640	0.582	+0.058
accumulator	JavaScript→Python	0.835	0.840	0.506	0.513	-0.008
accumulator	PHP→Python	0.831	0.809	0.554	0.424	+0.130
accumulator	Python→JavaScript	0.858	0.866	0.600	0.467	+0.133
accumulator	Python→PHP	0.865	0.854	0.476	0.446	+0.030
index_key	JavaScript→PHP	0.881	0.812	0.696	0.513	+0.183
index_key	PHP→JavaScript	0.935	0.824	0.659	0.568	+0.092
index_key	JavaScript→Python	0.875	0.805	0.661	0.556	+0.105
index_key	PHP→Python	0.858	0.746	0.594	0.370	+0.224
index_key	Python→JavaScript	0.879	0.862	0.672	0.520	+0.151
index_key	Python→PHP	0.878	0.828	0.576	0.328	+0.249
iterator	JavaScript→PHP	0.901	0.950	0.563	0.494	+0.069
iterator	PHP→JavaScript	0.980	0.859	0.668	0.491	+0.177
iterator	JavaScript→Python	0.577	0.836	0.523	0.473	+0.050
iterator	PHP→Python	0.548	0.780	0.517	0.514	+0.003
iterator	Python→JavaScript	0.830	0.966	0.597	0.476	+0.121
iterator	Python→PHP	0.810	0.885	0.556	0.477	+0.080

Table 7: Problem-clustered paired differences between the context-pooled probe and the surface comparator. Negative values favor the surface comparator.

role	pair	Δ	CI low	CI high
accumulator	JavaScript→PHP	-0.315	-0.403	-0.214
accumulator	JavaScript→Python	-0.253	-0.333	-0.178
accumulator	PHP→JavaScript	-0.302	-0.405	-0.211
accumulator	PHP→Python	-0.177	-0.269	-0.083
accumulator	Python→JavaScript	-0.243	-0.335	-0.150
accumulator	Python→PHP	-0.337	-0.443	-0.196
index_key	JavaScript→PHP	-0.069	-0.177	+0.034
index_key	JavaScript→Python	-0.159	-0.306	-0.051
index_key	PHP→JavaScript	-0.131	-0.240	-0.039
index_key	PHP→Python	-0.158	-0.297	-0.023
index_key	Python→JavaScript	-0.114	-0.200	-0.037
index_key	Python→PHP	-0.318	-0.410	-0.239
iterator	JavaScript→PHP	-0.346	-0.507	+0.000
iterator	JavaScript→Python	-0.135	-0.301	-0.031
iterator	PHP→JavaScript	-0.428	-0.508	-0.001
iterator	PHP→Python	-0.074	-0.225	+0.001
iterator	Python→JavaScript	-0.162	-0.392	+0.018
iterator	Python→PHP	-0.190	-0.346	-0.001

Table 8: Problem-clustered uncertainty for the aggregate boundary contrasts. The effect is Python-transfer minus JavaScript $\leftrightarrow$ PHP macro-F1. The difference-in-differences subtracts the untrained boundary from the trained boundary. Intervals resample problem identifiers and are conditional on the committed model runs, including one random weight initialization.

estimand	estimate	95% CI
span-pooled trained boundary	-0.012	[-0.034, +0.020]
occurrence-excluded trained boundary	-0.060	[-0.088, -0.030]
occurrence-excluded untrained boundary	-0.064	[-0.089, -0.039]
trained – untrained boundary	+0.003	[-0.027, +0.036]
training lift, close pair	+0.102	[+0.065, +0.132]
training lift, Python pairs	+0.106	[+0.081, +0.127]
boundary shift after exclusion	-0.048	[-0.088, -0.016]

Table 9: Rolewise occurrence-excluded effects and floor-adjusted transfer. Effects are Python minus close-pair macro-F1. Lift subtracts the matched untrained score. The final column is close-pair lift minus Python-transfer lift. These descriptive role summaries do not carry separate intervals.

role	trained effect	random effect	lift close	lift Python	Δ lift
accumulator	-0.062	-0.087	+0.046	+0.071	-0.025
index_key	-0.052	-0.097	+0.138	+0.182	-0.045
iterator	-0.067	-0.008	+0.123	+0.064	+0.060

Table 10: Per-cell macro-F1 for the StarCoder2-7B replication, on the same prediction items as the Qwen run. *Lift* subtracts the context-matched randomly initialized result from the context-pooled trained result.

role	pair	span	context	random	lift
accumulator	JavaScript $\rightarrow$ PHP	0.854	0.689	0.658	+0.030
accumulator	PHP $\rightarrow$ JavaScript	0.859	0.672	0.598	+0.075
accumulator	JavaScript $\rightarrow$ Python	0.844	0.516	0.516	-0.000
accumulator	PHP $\rightarrow$ Python	0.821	0.420	0.486	-0.066
accumulator	Python $\rightarrow$ JavaScript	0.881	0.636	0.561	+0.075
accumulator	Python $\rightarrow$ PHP	0.884	0.640	0.488	+0.152
index_key	JavaScript $\rightarrow$ PHP	0.846	0.768	0.573	+0.195
index_key	PHP $\rightarrow$ JavaScript	0.821	0.780	0.528	+0.252
index_key	JavaScript $\rightarrow$ Python	0.807	0.637	0.479	+0.158
index_key	PHP $\rightarrow$ Python	0.778	0.576	0.364	+0.211
index_key	Python $\rightarrow$ JavaScript	0.870	0.636	0.526	+0.110
index_key	Python $\rightarrow$ PHP	0.861	0.739	0.334	+0.405
iterator	JavaScript $\rightarrow$ PHP	0.946	0.767	0.523	+0.244
iterator	PHP $\rightarrow$ JavaScript	0.934	0.726	0.434	+0.292
iterator	JavaScript $\rightarrow$ Python	0.880	0.529	0.476	+0.053
iterator	PHP $\rightarrow$ Python	0.806	0.498	0.464	+0.033
iterator	Python $\rightarrow$ JavaScript	0.967	0.637	0.475	+0.162
iterator	Python $\rightarrow$ PHP	0.960	0.571	0.500	+0.071

Table 11: Boundary contrasts for the StarCoder2-7B replication, same estimator and resampling as the Qwen table. Conditional on one random weight initialization.

estimand	estimate	95% CI
span-pooled trained boundary	-0.014	[-0.036, +0.018]
occurrence-excluded trained boundary	-0.147	[-0.186, -0.092]
occurrence-excluded untrained boundary	-0.080	[-0.110, -0.049]
trained – untrained boundary	-0.068	[-0.114, -0.009]
training lift, close pair	+0.181	[+0.118, +0.227]
training lift, Python pairs	+0.114	[+0.086, +0.136]
boundary shift after exclusion	-0.134	[-0.168, -0.086]

Table 12: Qwen2.5-Coder-1.5B boundary contrasts recomputed against the second untrained weight initialization. Trained conditions are unchanged. Only the floor differs from the main table.

estimand	estimate	95% CI
span-pooled trained boundary	-0.012	[-0.034, +0.020]
occurrence-excluded trained boundary	-0.060	[-0.088, -0.030]
occurrence-excluded untrained boundary	-0.035	[-0.062, -0.008]
trained — untrained boundary	-0.025	[-0.061, +0.013]
training lift, close pair	+0.109	[+0.067, +0.145]
training lift, Python pairs	+0.084	[+0.058, +0.107]
boundary shift after exclusion	-0.048	[-0.088, -0.016]

Table 13: Aggregate diagnostics retained in the intersection-sample masked-probe export. Surface counts are test occurrences. Probe counts are shared source–target problems. Surface rows have no in-domain or seed fields. The untrained condition has two probe seeds versus five for the trained conditions.

condition	transfer	in-domain	shuffled source	seeds	cell population
surface window	0.846	N/A	0.476	N/A	2,617–3,138 occ.
span-pooled trained	0.844	0.898	0.468	5	374–406 problems
context-pooled trained	0.589	0.771	0.448	5	374–406 problems
context-pooled untrained	0.485	0.623	0.465	2	374–406 problems

234 D Renaming and formatting audits

Table 14: Uncapped renaming audit for Python-source transfer. C1, C2, and C4 are the committed identifier-renaming conditions. These samples differ slightly from the frozen main analysis and are reported as a robustness audit rather than pooled with it.

role	condition	target	n_{test}	name	surface	majority
accumulator	C1	JavaScript	12,746	0.396	0.790	0.423
accumulator	C1	PHP	3,517	0.222	0.768	0.417
accumulator	C2	JavaScript	12,746	0.410	0.789	0.423
accumulator	C2	PHP	3,517	0.473	0.786	0.417
accumulator	C4	JavaScript	12,746	0.389	0.793	0.423
accumulator	C4	PHP	3,517	0.565	0.817	0.417
index_key	C1	JavaScript	12,746	0.528	0.845	0.443
index_key	C1	PHP	3,517	0.599	0.880	0.439
index_key	C2	JavaScript	12,746	0.506	0.871	0.443
index_key	C2	PHP	3,517	0.626	0.879	0.439
index_key	C4	JavaScript	12,746	0.523	0.878	0.443
index_key	C4	PHP	3,517	0.509	0.879	0.439
iterator	C1	JavaScript	12,746	0.512	0.662	0.491
iterator	C1	PHP	3,517	0.443	0.650	0.489
iterator	C2	JavaScript	12,746	0.449	0.555	0.491
iterator	C2	PHP	3,517	0.337	0.538	0.489
iterator	C4	JavaScript	12,746	0.468	0.624	0.491
iterator	C4	PHP	3,517	0.431	0.630	0.489

Table 15: Whitespace-normalisation audit for the three decisive iterator surface transfers. Scores are unchanged at the displayed precision.

pair	raw	normalised	Δ
PHP→JavaScript	0.9654	0.9654	+0.0000
PHP→Python	0.5756	0.5756	+0.0000
Python→JavaScript	0.7901	0.7901	+0.0000

235 E Transfer-mechanism diagnostics

Table 16: Transfer mechanism for the iterator role. *Surviving* is the share of the source classifier’s discriminative mass realised in the target. *Flip* is the share of that mass whose sign reverses. Surviving mass is uncorrelated with transfer. Flipped mass predicts it almost exactly. Source-weighted variants are given for robustness.

pair	F1	surviving	flip	agree	flip (src-wt.)
PHP→JavaScript	0.965	0.735	0.072	+0.895	0.072
JavaScript→PHP	0.937	0.698	0.052	+0.945	0.068
Python→JavaScript	0.790	0.862	0.178	+0.695	0.211
JavaScript→Python	0.774	0.757	0.232	+0.811	0.241
Python→PHP	0.741	0.696	0.280	+0.528	0.367
PHP→Python	0.576	0.697	0.366	+0.476	0.346

Table 17: Masking an entire syntactic character class on both sides of the transfer. No class accounts for more than 0.021 of the 0.39 gap, so the realignment is distributed rather than carried by block delimiters or statement terminators.

pair	masked class	macro-F1	Δ
PHP→JavaScript	terminator	0.9496	-0.0159
PHP→JavaScript	brace	0.9645	-0.0010
PHP→JavaScript	bracket	0.9711	+0.0057
PHP→JavaScript	paren	0.9609	-0.0045
PHP→JavaScript	operator	0.9441	-0.0213
PHP→Python	terminator	0.5786	+0.0030
PHP→Python	brace	0.5750	-0.0006
PHP→Python	bracket	0.5733	-0.0023
PHP→Python	paren	0.5947	+0.0191
PHP→Python	operator	0.5801	+0.0045

236 F Language-geometry controls

237 **The typing axis is not privileged.** Probing the same activations under every alternative 4-versus-3
238 partition of the seven languages ($\binom{7}{4} = 35$ groupings minus the real one, so 34 controls) gives
239 best-layer macro-F1 0.9959 on average (min 0.9918, max 1.0000), against 1.0000 for the true
240 static/dynamic split. 5 of the 34 controls match or exceed it. Any grouping of these languages is
241 near-perfectly decodable, so the decodability of the typing split is evidence that language identity is
242 encoded, not that type discipline is. At layer 4, where $|r|$ peaks at 0.941, the remaining components
243 carry almost none of the label. PC2 $r = -0.014$, PC3 $r = +0.026$, PC4 $r = +0.171$.

Table 18: Point-biserial correlation between PC1 of last-token residual activations and a static/dynamic typing label, by layer, with PC1’s explained-variance ratio. The sign of r is arbitrary (PCA fixes the axis only up to reflection, and each layer is refit independently). Magnitude is what carries meaning.

layer	$ r $	EVR	layer	$ r $	EVR
0	0.883	0.481	15	0.894	0.309
1	0.898	0.447	16	0.882	0.303
2	0.914	0.391	17	0.874	0.296
3	0.939	0.373	18	0.873	0.289
4	0.941	0.370	19	0.873	0.299
5	0.862	0.369	20	0.903	0.291
6	0.828	0.361	21	0.886	0.276
7	0.844	0.342	22	0.859	0.265
8	0.896	0.339	23	0.842	0.252
9	0.919	0.333	24	0.812	0.242
10	0.880	0.327	25	0.774	0.253
11	0.893	0.302	26	0.694	0.255
12	0.917	0.292	27	0.732	0.237
13	0.904	0.314	28	0.576	0.242
14	0.898	0.309			

244 G Complete transfer heatmaps

245 **Reading the heatmaps.** Rows are source languages and columns are target languages. These plots
 246 retain the complete available transfer matrices, including cells excluded from matched-pair inference
 because problem overlap is too small.

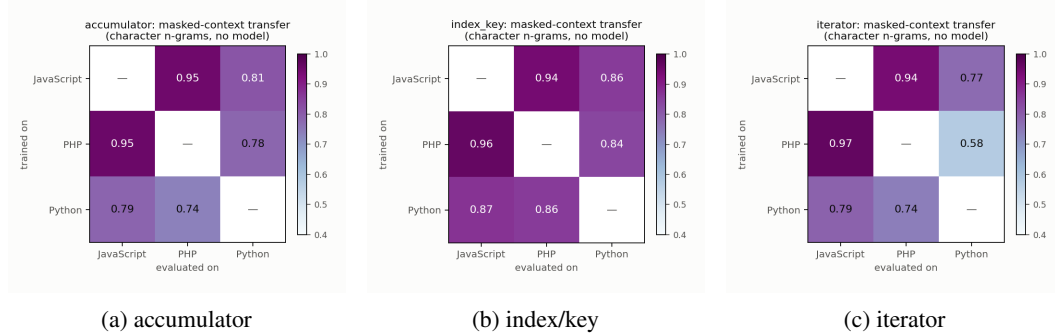


Figure 2: Available transfer matrices per role for Qwen2.5-Coder-1.5B. Cells without adequate matched problem overlap are descriptive only and are not used for the main boundary estimate.

247

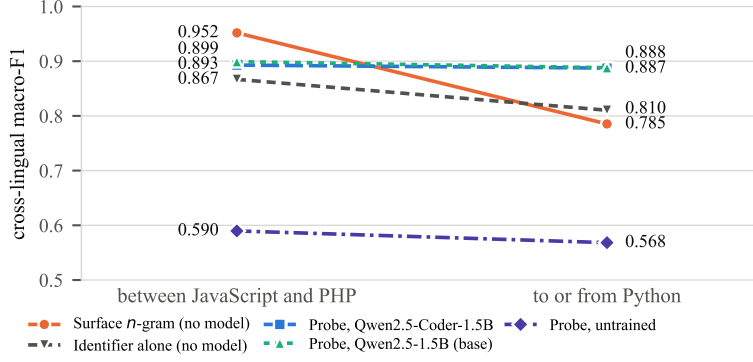


Figure 3: Span-pooled transfer contrast for the two trained models and the architecture-matched random-initialization floor. This capped-sample diagnostic is distinct from the intersection-sample context-pooled analysis.

248 H Reproducibility

249 Every figure is regenerated from committed result files, and a regression test recomputes the headline
 250 values from those files. Every condition in Table 1, both Qwen floor initializations, the StarCoder2
 251 floor initialization, and the prediction-level boundary analysis are committed per cell. Intervals
 252 resample stable problem identifiers once across all cells and conditions. Activation extraction is
 253 the only accelerator-dependent step. Probes use logistic regression on residual activations pooled
 254 over either the occurrence or the 16 tokens on each side. The surface baseline uses a `char_wb` tf-idf
 255 vectorizer with $n \in [2, 4]$ and source-language logistic regression. Store identities record model
 256 initialization and pooling so conditions cannot be mixed silently.

257 I Broader impacts

258 Positive effects are scientific. A span-pooled probe that looks language-invariant can be read as
 259 evidence that a code model represents roles independently of programming language. If such claims
 260 later inform multilingual coding tools, a readout-matched comparator reduces that overclaim.

261 Negative effects follow from the same scores. A flat transfer curve can be treated as invariance if the
 262 identifier remains in the readout. Role labels derived from public competitive-programming solutions
 263 could be reused to train name-based heuristics that recover the label from the identifier. No new
 264 generative model is released. The source programs were already public.

265 J Existing assets and licenses

266 Source programs come from XLCoST under Apache 2.0 [Zhu et al., 2022]. Qwen2.5-Coder-1.5B
 267 is used under Apache 2.0. StarCoder2-7B is used under the BigCode OpenRAIL-M license. Each
 268 release is cited by name and identifier. Derived role labels inherit Apache 2.0 from XLCoST.

269 K New assets

270 The new assets are the analysis scripts and committed result files that produce the reported tables,
 271 together with structurally labeled XLCoST variable-role data used as input. A dataset card documents
 272 fields, extractors, splits, and the parent license. An anonymized snapshot is at <https://anonymous.4open.science/r/CrossLanguageProbeInvariance2026>.
 273

274 NeurIPS Paper Checklist

275 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state the identifier-access confound, the matched context-pooled comparison, and the resulting limit on language-invariance claims. The scope is restricted to two models, three languages, and three roles.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The limitations paragraph covers model and language scope, role-specific coverage, extractor differences, random-seed counts, the mixed-sample span floor, residual identifier correlates, and the absence of causal evidence.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper reports no theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The setup gives the corpus restriction, problem-level split policy, feature restrictions, pooling rules, model identifiers, and source-validation layer selection. The reproducibility appendix records classifier settings and artifact provenance.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Analysis scripts, committed result files, and the labeled dataset are at <https://anonymous.4open.science/r/CrossLanguageProbeInvariance2026>. The reproducibility appendix records how every figure is regenerated from those files.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The setup specifies the frozen problem-hash split, fixed masked feature set, pooling rules, context width, and source-validation layer selection. The reproducibility appendix gives the tf-idf and logistic-regression configuration.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Problem-clustered intervals are reported for the surface contrast, paired cell comparisons, readout-induced boundary shift, and trained-minus-untrained boundary. These intervals resample problem identifiers and remain conditional on the committed model runs, including one random weight initialization.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: The paper identifies activation extraction as the accelerator-dependent step but does not report hardware, memory, or complete wall-clock costs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work analyses publicly released models on a public benchmark of competitive-programming solutions. It involves no human subjects, no personally identifying data, and no deployment.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix I states a scientific benefit for more conservative cross-language claims and a misuse path if a flat transfer curve is treated as invariance while the identifier remains in the readout.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The labeled dataset is a derivative of a public competitive-programming benchmark. It does not release a high-risk generative model or scraped media that would require extra safeguards.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Appendix J names Apache 2.0 for XLCOST and Qwen2.5-Coder-1.5B, and the BigCode OpenRAIL-M license for StarCoder2-7B. Derived labels inherit Apache 2.0. Each asset is cited.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Appendix K documents the labeled XLCoST variable-role data, the dataset card covering fields and the parent license, and committed analysis artifacts. The anonymized snapshot is at <https://anonymous.4open.science/r/CrossLanguageProbeInvariance2026>.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

588 Question: Does the paper describe potential risks incurred by study participants, whether
589 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
590 approvals (or an equivalent approval/review based on the requirements of your country or
591 institution) were obtained?

592 Answer: [N/A]

593 Justification: The paper involves no human subjects, so no IRB review was required.

594 Guidelines:

- 595 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
596 with human subjects.
- 597 • Depending on the country in which research is conducted, IRB approval (or equivalent)
598 may be required for any human subjects research. If you obtained IRB approval, you
599 should clearly state this in the paper.
- 600 • We recognize that the procedures for this may vary significantly between institutions
601 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
602 guidelines for their institution.
- 603 • For initial submissions, do not include any information that would break anonymity (if
604 applicable), such as the institution conducting the review.

605 16. Declaration of LLM usage

606 Question: Does the paper describe the usage of LLMs if it is an important, original, or
607 non-standard component of the core methods in this research? Note that if the LLM is used
608 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
609 scientific rigor, or originality of the research, declaration is not required.

610 Answer: [N/A]

611 Justification: Large language models are the object of study, not an important or non-standard
612 component used to develop the research method. Writing and editing assistance is exempt
613 under the question.

614 Guidelines:

- 615 • The answer [N/A] means that the core method development in this research does not
616 involve LLMs as any important, original, or non-standard components.
- 617 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
618 be described.